

SAFETY DATA SHEET

Page 1

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

IDENTITY
AWC® C-221

Section I Company and Product Identification

AMERICAN WATER CHEMICALS® 1802 CORPORATE CENTER LANE PLANT CITY, FL 33563 Telephone Number: (813)-246-5448	ChemTel phone Number: In the U.S.: 1-800-255-3924 International: 1-813-248-0585 Australia: 1-300-954-583 Brazil: 0-800-591-6042 China: 400-120-0751 India: 000-800-100-4086 Mexico: 800-099-0731
Date Prepared: 11/21/03	Date Revised: October, 2020

Trade Name	AWC C-221
Product Family	Membrane Preservative

Section II – Hazards Identification

Signal Word : DANGER Acute Toxicity: TOXIC TO REPRODUCTION (Fertility) - Category 1B TOXIC TO REPRODUCTION (Unborn child) - Category 1B Oral, Category 5 May be harmful if swallowed. Dermal, Category 4 Harmful in contact with skin Inhalation Category 4 Harmful if inhaled.	
GHS Hazard Phrases	May damage fertility or the unborn child. H303: May be harmful if swallowed. H332: Harmful if inhaled H312: Harmful in contact with skin.
GHS Precaution Phrases	P103: Read label before use P261: Do not breathe dust/fume/gas/mist/vapours/spray. P264: Wash thoroughly after handling P270: Do not eat drink or smoke while using this product.
GHS Response Phrases	P311: Call a poison center/doctor/...if you feel unwell
GHS Storage and Disposal Phrases	Please refer to section 7 for storage and section 13 for Disposal information

SAFETY DATA SHEET

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

Route(s) of Entry:	Inhalation? Yes	Skin? YES	Ingestion? No
Health Hazards (Effects of Acute and Chronic Overexposure)			
Inhalation: Sore throat, shortness of breath coughing, and congestion.			
Eye Contact: Contact with eyes will cause irritation.			
Skin Contact: Skin irritant. Symptoms include itching, redness and pain. May be harmful if absorbed through the skin. May cause an allergic reaction in certain individuals.			
Ingestion (Swallowing): Ingestion is not a likely route of exposure to this product. May cause irritation to the mucus membranes.			

Section III - Hazard Ingredients/Composition Information

Hazardous Components (Chemical Name)	CAS #	%
Sodium metabisulfite	7681-57-4	> 30%
Boric Acid	10043-35-3	> 30%

Section IV – First Aid Procedures

Inhalation: If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention.
Eye Contact: Check for and remove any contact lenses. In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Cold water may be used. Get medical attention immediately.
Skin Contact: In case of contact, immediately flush skin with plenty of water. Cover the irritated skin with an emollient. Remove contaminated clothing and shoes. Cold water may be used. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.
Ingestion (Swallowing): Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If large quantities of this material are swallowed, call a physician immediately. Loosen tight clothing such as a collar, tie, belt or waistband.

Section V - Fire Fighting Measures

Flash Point (Method Used) None	Flammable Limits: NE
Extinguishing Media: Dry Powder is recommended ☐	
Auto ignition temp: Noncombustible.	
Special Fire Fighting Procedures and Protective Equipment: Because fire may produce toxic thermal decomposition products, wear a self-contained breathing apparatus (SCBA) with a full face piece operated in pressure-demand or positive-pressure mode.	
Unusual Fire and Explosion Hazard: May release hazardous gas with fire or water.	
Hazardous Decomposition or Combustion Byproducts: When heated to decomposition it emits toxic fumes of Sox, Na2O. Contact with acids liberates toxic gas, sulfur dioxide. Slowly oxidized to the sulfate on exposure to air and moisture. Corrosive to aluminum in aqueous solution.	

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

Section VI – Accidental Release Measures

Steps to Be Taken in Case Material is Released or Spilled: Small Spill: Vacuum or sweep up material and place into a suitable disposal container. Clean up spills immediately, observing precautions in the Protective Equipment section. Avoid generating dusty conditions. Provide ventilation. Finish cleaning by spreading water on the contaminated surface and dispose of according to local and regional authority Requirements.

Large Spill: Use a shovel to put the material into a convenient waste disposal container. Finish cleaning by spreading water on the contaminated surface and allow to evacuate through the sanitary system. Be careful that the product is not present at a concentration level above TLV. Check TLV on the MSDS and with local authorities.

Additional Information: Planning ahead is essential for handling spills. Proper equipment and trained employees should be readily available to correct a spill situation. Prevent entry to sewers and public waters. Notify authorities if material enters sewers and public waters. Avoid release to the environment.

Section VII - Handling and Storage

Precautions to Be taken in Handling (General):	Keep locked up. Do not ingest. Do not breathe dust. Avoid contact with skin. Wear suitable protective clothing. In case of insufficient ventilation, wear suitable respiratory equipment. If ingested, seek medical advice immediately and show the container or the label. Keep away from incompatibles such as oxidizing agents, acids.
Precautions for safe storage and any incompatibilities	Keep container tightly closed. Keep container in a cool, well-ventilated area. Moisture sensitive. Air Sensitive. Keep away from direct sunlight.

Section VIII – Exposure Controls and Personal Protection

Components (Specific Chemical Identity; Common Name(s))	OSHA PEL	TWA8 ACGIH	STEL ACGIH	Other Limits Recommended
Sodium metabisulfite	None	5 (mg/m ³) TWA	None	NA
Sulfur dioxide (SO ₂ gas)	5 ppm TWA; 13 mg/m ³ TWA	2 ppm TWA; 5 mg/m ³ TWA 100 ppm IDLH	2 ppm TWA; 5 ppm STEL	

Primary Route of Exposure: None known	
Target Organs: None Known	
Respiratory Protection: For many conditions, no respiratory protection may be needed; however in dusty or unknown atmospheres or when exposures exceed limit values, use NIOSH-approved dust respirator.	
Skin Protection: For brief contact, few precautions other than clean body covering clothing should be needed. When prolonged or frequently repeated contact could occur, use protective full body clothing such as PVC or rubber, impervious to this material.	
Protective Gloves: Protective gloves made of PVC, neoprene or rubber.	Eye Protection: Wear dustproof goggles.
Other Protective Clothing or Equipment: Eye wash facility and safety shower in immediate area.	

SAFETY DATA SHEET

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

Section IX - Physical/Chemical Characteristics

Boiling Point: N/A	Bulk Density (H ₂ O = 1): 1.4 g/cm ³
Viscosity : (Brookfield) NE	Specific Gravity (H ₂ O = 1):N/A
Solubility in Water: Soluble	pH: 1 % Solution 7-9
Appearance and Odor: white to yellow solid with sulfur dioxide odor.	

NR: NOT REQUIRED, NE: NOT ESTABLISHED, NA: NOT APPLICABLE

Section X – Stability & Reactivity Data

Stability: Stable under normal conditions
Conditions to avoid: Reactive with oxidizing agents.
Incompatibility (Materials to Avoid): Corrosive to aluminum in aqueous solution Moisture sensitive Air sensitive. Slowly oxidizes to sodium sulfate upon exposure to air and moisture. Incompatible with sodium nitrite
Hazardous Decomposition Products: Oxides of sulfur, toxic fumes of sodium oxide
Hazardous Polymerization: Does not occur
Condition to avoid: None known
Additional Information: none

Section XI– Toxicological Information

RTECS#: CAS# 7681-57-4: UX8225000 CAS# 7446-09-5: WS4550000 LD50/LC50: CAS# 7681-57-4 (sodium metabisulfite): Dermal, guinea pig: LD50 = >1 gm/kg; Draize test, rabbit, eye: 100 mg/24H Mild; Draize test, rabbit, skin: 500 mg; Oral, rat: LD50 = 1131 mg/kg; Skin, rat: LD50 = >2 gm/kg; . CAS# 7446-09-5 (sulfur dioxide): Draize test, rabbit, eye: 6 ppm/32D Mild; Inhalation, mouse: LC50 = 3000 ppm/30M; Inhalation, rat: LC50 = 2520 ppm/1H; Inhalation, rat: LC50 = 2168 mg/m ³ ; . Carcinogenicity: CAS# 7681-57-4: Not listed by ACGIH, IARC, NTP, or CA Prop 65. CAS# 7446-09-5: Not listed by ACGIH, IARC, NTP, or CA Prop 65. Epidemiology: Two cases of occupational asthma in laundry workers exposed to sodium metabisulfite were reported. Sodium metabisulfite may be considered to be the anhydride of sodium bisulfite and is the chief constituent of commercial dry sodium bisulfite. Teratogenicity: No information found Reproductive Effects: See actual entry in RTECS for complete information. Mutagenicity: See actual entry in RTECS for complete information. Neurotoxicity: No information found Chronic Effects on Humans: Mutagenic Effects: Mutagenic for bacteria and/or yeast. May cause damage to the following organs: upper respiratory tract, skin, eyes. Other Toxic Effects on Humans: Hazardous in case of skin contact (irritant), of ingestion, of inhalation. Slightly hazardous in case of skin contact (permeator). Special Remarks on Toxicity to Animals: Not available. Special Remarks on Chronic Effects on Humans: May affect genetic material (mutagenic) based on animal test data.

SAFETY DATA SHEET

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

May cause adverse reproductive effects based on animal test data.
Special Remarks on other Toxic Effects on Humans:
Acute Potential Health Effects:
Skin: May cause skin irritation.
Eyes: May cause eye irritation.
Inhalation: May cause respiratory tract irritation with coughing and wheezing.
Ingestion: May be harmful if swallowed. May cause gastrointestinal tract irritation with abdominal pain, Nausea, vomiting, diarrhea, violent colic, and possible gastric hemorrhaging. May affect behavior/central nervous System and cause central nervous system depression/seizures. It may also affect the cardiovascular system (hypotension, tachycardia, cardiovascular collapse), Ingestion of sulfite compounds may cause a severe allergic Reaction (anaphylactoid symptoms) in sensitive individuals and some asthmatics.
Chronic Potential Health Effects:
Skin: Prolonged or repeated skin contact may cause allergic dermatitis.
Ingestion: Prolonged or repeated ingestion may affect the liver, urinary system, and metabolism (weight loss). Future exposures may also cause asthma like allergy with coughing, shortness of breath, wheezing and/or chest Tightness.
Inhalation: Prolonged or repeated inhalation may irritate the lungs, may cause bronchitis to develop with cough, Phlegm and/or shortness of breath.

Section XII – Ecological Information

Ecotoxicity: Not available.
Products of Biodegradation:
Possibly hazardous short term degradation products are not likely. However, long term degradation products may Arise.
Toxicity of the Products of Biodegradation: The products of degradation are less toxic than the product itself.
Special Remarks on the Products of Biodegradation: Not available
Persistence and degradability:
BOD-5 (mg/g): 0
BOD-28 (mg/g): 0
COD (mg/g): 62 (estimated based on similar materials)
TOC (mg/g): 0

Section XIII - Disposal Consideration

Waste Disposal Method: Dispose of in approved waste facility operated by authorized contractor in compliance with federal, state and local regulations

Section XIV - Transport information

D.O.T. Proper Shipping Name: N/A	D.O.T. Hazard Class: N/A D.O.T. LABEL: N/A
ID No: N/A	Packing Group: N/A

SAFETY DATA SHEET

AMERICAN WATER CHEMICALS®
1802 CORPORATE CENTER LANE
PLANT CITY, FL 33563

Section XV - Regulatory information

Federal and State Regulations:

Connecticut hazardous material survey.: Sodium metabisulfite
Illinois toxic substances disclosure to employee act: Sodium metabisulfite
Rhode Island RTK hazardous substances: Sodium metabisulfite
Pennsylvania RTK: Sodium metabisulfite
Minnesota: Sodium metabisulfite
Massachusetts RTK: Sodium metabisulfite
New Jersey: Sodium metabisulfite
California Director's List of Hazardous Substances: Sodium metabisulfite
TSCA 8(b) inventory: Sodium metabisulfite
TSCA 4(a) ITC priority list: Sodium metabisulfite
TSCA 8(a) PAIR: Sodium metabisulfite
TSCA 8(d) H and S data reporting: Sodium metabisulfite: effective: 1/26/94; sunset: 6/30/98

Other Regulations:

OSHA: Hazardous by definition of Hazard Communication Standard (29 CFR 1910.1200).
EINECS: This product is on the European Inventory of Existing Commercial Chemical Substances.

Other Classifications:

WHMIS (Canada): CLASS D-2A: Material causing other toxic effects (VERY TOXIC).

DSCL (EEC):

HMIS (U.S.A.):

Health Hazard: 2

Fire Hazard: 0

Reactivity: 0

Personal Protection: E

National Fire Protection Association (U.S.A.):

Health: 2

Flammability: 0

Reactivity: 0

Specific hazard:

Protective Equipment:

Section XVI - Other information

NFPA RATINGS:

Health	Flammability	Reactivity
2	0	1

The data contained in this material safety data sheet has been prepared based upon an evaluation of the ingredients contained in the product, their concentrations in the product and potential interactions. The information is offered in good faith and is believed to be accurate. It is furnished to the customer who is urged to study it carefully to become aware of hazards, if any, in the storage, handling, use and disposal of the product; and to insure his employees are properly informed and advised of all safety precautions required.